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ANSWER

```
n=int(input("enter a number:"))
s=0
for i in str(n):
    s+=int(i)
if n % s ==0:
    print("Harshad number")
else:
    print("not a Harshad number")
print("\n Harshad numbers from 1 to 500:")
for n in range(1,501):
    s=0
    for i in str(n):
        s+=int(i)
    if n % s ==0:
        print(n, end=" ")
```

Q2. GCD

CODE

```
a=int(input("enter first:"))
b=int(input("enter second:"))
while b!=0:
    temp=a
    a=b
    b=temp%b
print("GCD is",a)
```

INPUT

24
18

OUTPUT enter first:enter second:GCD is 6

Q3. sum to a target value

ANSWER

```
def find_pairs(list,target):
    seen = set();pairs = []
    for num in list:
        complement=target-num
        if complement in seen:
            pairs.append((complement,num))
        seen.add(num)
    return pairs
data =[int(x) for x in input ("enter numbers:").split()]
t=int(input("target:"))
result=find_pairs(data,t)
print("pairs:",result)
```

Q4. Bitwise Operators

ANSWER

```
a=0b1101
b=0o25
c=0x1A

result=(a & b)|c
```

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Student Submissions — Analytical

Q5. binary search

CODE

```
def rotated_binary_search(arr, target):
    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == target:
            return mid

        if arr[low] <= arr[mid]:
            if arr[low] <= target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1
        else:
            if arr[mid] < target <= arr[high]:
                low = mid + 1
            else:
                high = mid - 1

    return -1

data = [4, 5, 6, 7, 0, 1, 2]
t = int(input("Target: "))
result = rotated_binary_search(data, t)
print("Found at:", result)
```

OUTPUT Target: Found at: 4